

1) we have n obs.

$y_i \rightarrow$ response

$x_{i1}, x_{i2}, \dots, x_{ip} \rightarrow$ predictors

Model:- $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$

$$\beta = (\beta_0, \beta_1, \dots, \beta_p)^T \in \mathbb{R}^{p+1}$$

$$X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times (p+1)}$$

$$y = (y_1, \dots, y_n)^T \in \mathbb{R}^n, \epsilon = (\epsilon_1, \dots, \epsilon_n)^T$$

$$y = X\beta + \epsilon$$

assumption:-

a) linearity :- $y = X\beta + \epsilon$

b) Homoscedasticity $E(\epsilon | X) = 0$

c) No auto correlation: $\text{Cov}(\epsilon_i, \epsilon_j | X) = 0$ for $i \neq j$

2) residual vector $r(\beta) = y - X\beta$

$$SSE(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T (y - X\beta)$$

MSE objective

$$= \frac{1}{n} (y - X\beta)^T (y - X\beta)$$

$$MSE(\beta) = \frac{1}{n} (y^T y - 2\beta^T X^T y) + \beta^T X^T X \beta$$

3) start with $\rightarrow J(\beta) = \frac{1}{n} (y - X\beta)^T (y - X\beta)$

$$= \frac{1}{n} (y^T y - 2\beta^T X^T y + \beta^T X^T X \beta)$$

Diff wrt β

$$\nabla_{\beta} J(\beta) = \frac{1}{n} (-2X^T y + 2X^T X \beta)$$

$$= \frac{2}{n} X^T (X\beta - y) = 0$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

4) $X^T X$ is invertible, is

$\rightarrow$ No column of X is linear combination of other columns

$$\dim K(X) = p+1$$

$\therefore$ we require $n \geq p+1$

if predictors are collinear,

then there exists a vector a such that

$$Xa = 0$$

then $a^T X^T X a = \|Xa\|^2 = 0$

So $X^T X$ has non zero vector in its null space $\rightarrow$ determinant $= 0 \rightarrow$ singular
 $\downarrow$
 non invertible

5) define fitted values

$$\hat{y} = X\hat{\beta}$$

$$e = y - \hat{y}$$

from normal eqns:

$$X^T(y - X\hat{\beta}) = 0$$

$$X\hat{\beta} = \hat{y}$$

$$X^T(y - \hat{y}) = 0$$

$\therefore$ columns of X span the model subspace, $\text{col}(X)$

$X^T e = 0$ means residual e is orthogonal to every column of X .

$$\therefore y = \hat{y} + e \rightarrow \begin{aligned} \hat{y} &\in \text{col}(X) \\ e &\perp \text{col}(X) \end{aligned}$$

$\downarrow$
 $\therefore$ orthogonal projection of y onto $\text{col}(X)$

6) we derived $\nabla_{\beta} J(\beta)$ earlier

$$\nabla_{\beta} J(\beta) = \frac{1}{n} X^T (X\beta - y)$$

Batch gradient ~~descent~~ update
learning rate $\eta > 0$

$$\beta^{(t+1)} = \beta^{(t)} - \eta \frac{1}{n} X^T (X\beta - y)$$

$$12) E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Var} + \sigma^2$$

$$y = f(x) + \epsilon, \quad E(\epsilon) = 0, \quad \text{var}(\epsilon) = \sigma^2$$

let $\hat{f}(x)$ be a learned estimator

$$\begin{aligned} E[(y - \hat{f}(x))^2] &= E[(f(x) + \epsilon - \hat{f}(x))^2] \\ &= E[(f(x) - \hat{f}(x))^2] + 2E[\epsilon(f(x) - \hat{f}(x))] + E(\epsilon^2) \end{aligned}$$

as ϵ has mean 0
 ϵ independent of $\hat{f}(x)$ given x

$$= E[(f(x) - \hat{f}(x))^2] + \sigma^2$$

$$f(x) - \hat{f}(x) = (f(x) - E(\hat{f}(x))) + (E(\hat{f}(x)) - \hat{f}(x))$$

$$E[(f - \hat{f})^2] = \underbrace{(f - E(\hat{f}))^2}_{\text{Bias}} + \underbrace{E[(\hat{f} - E(\hat{f}))^2]}_{\text{var}}$$

$$= \text{Bias}^2 + \text{var} + \sigma^2$$

13)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$\text{fit } y = \alpha_0 + \alpha_1 x_1 + u$$

(a) slope regressing y on x_1 alone:

$$\alpha_1 = \frac{\text{Cov}(x_1, y)}{\text{Var}(x_1)}$$

$\text{Cov}(x_1, y)$:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$\text{Cov}(x_1, y) = \beta_1 \text{Cov}(x_1, x_1) + \beta_2 \text{Cov}(x_1, x_2) + \text{Cov}(x_1, \varepsilon)$$

$$\text{Cov}(x_1, \varepsilon) = 0$$

$$\text{Cov}(x_1, y) = \beta_1 \text{Var}(x_1) + \beta_2 \text{Cov}(x_1, x_2)$$

$$E(\alpha_1) = \beta_1 + \beta_2 \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)}$$

$$b) \quad E(\alpha_1) - \beta_1 = \frac{\beta_2 \text{Cov}(x_1, x_2)}{\text{Var}(x_1)}$$

if x_1 correlated with omitted x_2 & $\beta_2 \neq 0$
 then α_1 shifts.

(c) when does bias disappear?
 Bias is zero if any of these hold:

1) $\beta_2 = 0$

2) $\text{Cov}(x_1, x_2) = 0$

3)